EU compliance pack — PixelPine rev A

Product: USB-C (5 V, ≤ 0.1 W) decorative LED Christmas tree, 240 LEDs, no radio, no battery, not a toy. Manufacturer placing it on the market: **MachStuff s.r.o.**, Jehličková 240, 240 24 Stromečkov, Czech Republic, marekmach24@gmail.com (the address is made up for the school project). That makes us the "manufacturer" under EU law even though assembly is subcontracted in China.

> This is an engineering-student compliance plan, not legal advice. Before the first sale, confirm the applicability matrix with the test lab or a notified-body consultant (a one-hour call, typically free with a test order).

Applicability matrix

Legislation	Applies?	Why	What we do
EMC Directive 2014/30/EU	**Yes → CE**	Electrical apparatus that can emit or suffer electromagnetic disturbance (fast GPIO edges, exposed touch pad)	Internal production control (Module A). Test to **EN IEC 55015** (emissions, lighting equipment) and **EN 61547** (immunity, lighting). Test plan
RoHS 2011/65/EU + (EU) 2015/863	**Yes → CE**	Electrical & electronic equipment	Lead-free HASL finish (**not** leaded HASL), RoHS declarations from LCSC for every part, 1 XRF screen of a finished unit. Technical file per **EN IEC 63000**
Low Voltage Directive 2014/35/EU	No	Only covers 50–1000 V AC / 75–1500 V DC; we run on 5 V DC	Electrical safety is handled under GPSR instead
General Product Safety Regulation (EU) 2023/988 (GPSR)	**Yes**	Consumer product; the LVD doesn't cover it	Risk assessment, technical documentation, product traceability markings, safety instructions in the local language, a contact for safety issues. Technical file
WEEE Directive 2012/19/EU (CZ: Act 542/2020 Sb.)	**Yes**	EEE, category 5 (small equipment)	Crossed-out-bin symbol **on the product** (back silkscreen ✓). Register as a producer and join a take-back scheme (e.g. ASEKOL, ELEKTROWIN, REMA) in every country we sell to
Packaging & Packaging Waste Regulation (EU) 2025/40 (PPWR)	**Yes, from 12 Aug 2026**	We place packaging on the market	Minimal, recyclable kraft box, no plastic window; EPR registration (CZ: EKO-KOM). Harmonised sorting labels come later (2028+)
REACH 1907/2006	**Yes**	Articles	Art. 33 SVHC check via supplier declarations; answer consumer SVHC requests within 45 days
Toy Safety Directive 2009/48/EC / Regulation (EU) 2025/2509	**No**	Annex I of both lists **Christmas decorations** as products *not* considered toys	Label "Not a toy. Decoration for indoor use." Design and marketing must not target children
Ecodesign for light sources (EU) 2019/2020	No	The regulation covers light sources of **60–82 000 lm**; the whole tree emits ≈ **2 lm** (calculation)	Keep the calculation in the technical file
Energy labelling (EU) 2019/2015	No	Same 60 lm scope as above	—
Radio Equipment Directive 2014/53/EU	No	No intentional radio	If a BLE variant ever ships: RED, EN 300 328, EN 18031 cybersecurity
Common charger (EU) 2022/2380	No	Amends RED, radio equipment only. We still use USB-C, in the spirit of it	—
Batteries Regulation (EU) 2023/1542	No	No battery	—
Product Liability Directive (EU) 2024/2853	Yes (liability)	Covers any product	Product-liability insurance (~€150–300/yr for a small seller); keep the technical file for 10 years

Legislation	Applies?	Why	What we do
Photobiological safety (EN 62471)	Hazard check only	Under GPSR, LED products are checked for blue-light / retinal hazard	≈ 2 lm of 2700 K light is trivially **Exempt Group**; document the reasoning, no test needed

Result: CE marking covers **EMC + RoHS** only. GPSR, WEEE, PPWR and REACH are "must do" obligations without their own mark (WEEE has the bin symbol).

Design decisions that were made *for* compliance

Decision	Regulation it serves	Cost
Lead-free HASL surface finish	RoHS	≈ +$0.02/board
R19 1 kΩ in series with the exposed star touch pad	EN 61547 ESD immunity (±4 kV contact / ±8 kV air)	$0.004
100 Ω on every charlieplex line + GPIO speed set to 2 MHz (firmware)	EN IEC 55015 radiated emissions: slow edges	€0, the resistors are needed anyway
No radio, no USB data	Avoids RED + cybersecurity testing	saves €3–5 k
Rounded board outline, no sharp tips	GPSR mechanical hazards (cuts, pokes)	€0, and it's cuter
Traceability marking on the back silkscreen (model, batch, manufacturer, address, e-mail)	GPSR Art. 9	€0
CE + WEEE logos on the back silkscreen	EMC/RoHS marking, WEEE Art. 14	€0

Roadmap and budget

#	Step	When	Cost (EUR)
1	Freeze rev A, route the board	T−8 weeks	—
2	Order 100-pc pilot; collect LCSC RoHS/REACH declarations for all 8 part numbers	T−7 w	in pilot cost
3	Write risk assessment + technical file draft (technical-file.md)	T−6 w	own time
4	**EMC pre-scan** at a university lab (near-field probe + spectrum analyser, ESD gun)	T−5 w	0–300
5	**Accredited EMC test** EN IEC 55015 + EN 61547 (e.g. EZÚ Praha, ITC Zlín, or a DE/AT lab)	T−4 w	1 500–2 500
6	XRF RoHS screen of one finished unit	T−4 w	~400
7	Sign the EU Declaration of Conformity (template)	T−3 w	—
8	WEEE producer registration + take-back scheme; EKO-KOM packaging contract	T−3 w	~300/yr + fees per kg
9	Print box and leaflet (texts)	T−3 w	in packaging cost
10	Product-liability insurance	T−2 w	150–300/yr
11	Start selling. For online shops: show the manufacturer's contact, product photo, warnings and identifiers on the listing (GPSR Art. 19)	T	—

Budget used in cost-analysis.md: **€2 700** (steps 5, 6, 8), plus insurance.

Files in this folder

- test-plan.md: EMC test list, setup, limits and risk points
- technical-file.md: technical documentation index + GPSR risk assessment
- declaration-of-conformity.md: EU DoC template, ready to fill in
- labelling-and-instructions.md: on-product marking, box texts, leaflet (EN + CZ)